1 Versió 1 (la més recent)

Pregunta 1

Resposta desada

Puntuat sobre 10,00

Consider the mesh called `meshEscaire2foratsTriang`. Let $p_0 = (-1.4, 3.4)$, and assign the following *distance value* to each node v:

$$f(v) = \|v - p_0\|,$$

that is, $f(v)$ is the distance from v to p_0.
Hint: In Matlab, the distance from v to p_0 is computed with `norm(v-p0)`.

(a) (2 points) The farthest (most distant) node from p_0 is node number:

- ○ 83
- ○ 85
- ○ 84
- • 82
- ○ Empty answer (no penalty)
- ○ 81

Hint: The command `[M,I]=max(A)` returns the maximum element `M` and its index `I`.

(b) (4 points) Interpolate the function f at the point $p = (5.1, 2.9)$. The interpolated value is:

- ○ 6.4413e+00
- ○ Empty answer (no penalty)
- ○ 6.4478e+00
- ○ 6.5848e+00
- • 6.5196e+00

(c) (2 points) Given a triangular element, its barycenter g has barycentric coordinates $(1/3, 1/3, 1/3)$. Find the barycenter of element number 7 in the mesh:

- • $g = (5.1556e+00, 6.9048e-01)$
- ○ $g = (4.6400e+00, 6.2143e-01)$
- ○ $g = (5.6196e+00, 7.5262e-01)$
- ○ $g = (4.5369e+00, 6.0762e-01)$
- ○ Empty answer (no penalty)

(d) (2 points) Given a triangular element with barycenter g, we define its *interpolation error* as the difference between the interpolated value of f at g and the exact value of f at g. The interpolation error for element number 7 in the mesh is:

- ○ 3.9831e-03
- ○ 4.0937e-03
- ○ 4.0568e-03
- ○ Empty answer (no penalty)
- • 3.6880e-03

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› Opcions de previsualització

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